Fundamentals of Machine Learning
Hard Mock Midterm: Solutions

Every new result proved from the course's tools, every number computed, and the line that connects it back to a lecture

Fundamentals of Machine Learning

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1. Amount-dependent costs and noisy labels

(a) Conditional risks: $R(\text{block} \mid x) = c\,(1-\eta)$, $R(\text{allow} \mid x) = A(x)\,\eta$. Block iff $c(1-\eta) < A(x)\eta$ iff $\eta > t(x) = \dfrac{c}{c + A(x)}$. For 20 euros: $5/25 = 0.20$. For 500 euros: $5/505 = 0.0099$. The blocked set is $\{x : \eta(x) > c/(c + A(x))\}$: two transactions with the same η get different decisions, so no single cut on the score describes the system. A reported "threshold" is meaningless; one must report the rule $t(x)$ or the expected cost.
(b) Condition on the true class (the flips do not depend on x given Y): $\tilde{\eta} = \mathbb{P}(\tilde{Y} = 1 \mid Y = 1)\,\eta + \mathbb{P}(\tilde{Y} = 1 \mid Y = 0)(1-\eta) = (1-\rho_1)\eta + \rho_0(1-\eta) = \rho_0 + (1-\rho_0-\rho_1)\eta$.
(c) Cross-entropy is proper, so the model estimates $\tilde{\eta}$. The map $\eta \mapsto \tilde{\eta}$ is increasing (part (d)), so $\eta > t \iff \tilde{\eta} > \tilde{t} = \rho_0 + (1-\rho_0-\rho_1)t = 0.01 + 0.69 \times 0.2 = 0.148$. Using 0.2 on the noisy model would miss frauds whose clean posterior lies between 0.2 and $(0.2 - 0.01)/0.69 = 0.275$.
(d) $\tilde{\eta}$ is an affine function of η with slope $1 - \rho_0 - \rho_1 > 0$, so $\eta(x) > \eta(x') \iff \tilde{\eta}(x) > \tilde{\eta}(x')$. Everything that depends only on the order of the scores is unaffected: the ROC curve and its AUC (computed against the true labels), the set of decisions achievable by *some* threshold. Calibration, the meaning of a given threshold, and every metric at a fixed threshold are affected. With $\rho_0 + \rho_1 = 1$ the slope is 0: $\tilde{\eta} = \rho_0$ for every x, the recorded labels are independent of the features, and nothing can be learned.
(e) With $\rho_0 = \rho_1 = \rho$: $\tilde{t} = \rho + (1-2\rho)\frac{1}{2} = \frac{1}{2}$, so the zero–one rule "$\tilde{\eta} > \frac{1}{2}$" is exactly "$\eta > \frac{1}{2}$". For $t \neq \frac{1}{2}$, $\tilde{t} = \rho + (1-2\rho)t \neq t$ (it is pulled toward $\frac{1}{2}$), so applying the clean cost threshold to the noisy model is wrong; symmetric noise is harmless only at the one threshold it maps to itself.

2. Proper losses

(a) $\mathbb{E}[(Y-q)^2 \mid x] = \eta(1-q)^2 + (1-\eta)q^2 = \eta - 2\eta q + q^2 = \eta(1-\eta) + (\eta^2 - 2\eta q + q^2) = \eta(1-\eta) + (\eta - q)^2$. The first term does not depend on q, the second is zero only at $q = \eta$: proper. $\eta(1-\eta)$ is the irreducible noise (variance of a Bernoulli), $(\eta - q)^2$ the calibration error: the Lecture 3 decomposition for a binary target.
(b) $\mathbb{E}[|Y-q| \mid x] = \eta(1-q) + (1-\eta)q = \eta + q(1-2\eta)$, linear in q. Its minimiser on $[0,1]$ is $q = 1$ if $\eta > \frac{1}{2}$ and $q = 0$ if $\eta < \frac{1}{2}$: the model outputs hard labels, never η (for $\eta = 0.3$: risk 0.3 at $q = 0$, 0.42 at $q = 0.3$, 0.7 at $q = 1$). Harmless for zero–one decisions, which only need the side of $\frac{1}{2}$. Fatal for the cost threshold 0.2: the true $\eta = 0.3 > 0.2$ says alert, but the model reports 0 and the flow is let through.
(c) $\mathbb{E}[-\log q_Y \mid x] = H(\eta) + \mathrm{KL}(\eta \| q) \geq H(\eta)$ with equality iff $q = \eta$ (Gibbs): proper.
(d) $\mathbb{E}[\ell_w \mid x] = -w_1\eta\log q - w_0(1-\eta)\log(1-q)$. Derivative in q: $-\frac{w_1\eta}{q} + \frac{w_0(1-\eta)}{1-q} = 0 \Rightarrow w_1\eta(1-q) = w_0(1-\eta)q \Rightarrow q^* = \frac{w_1\eta}{w_1\eta + w_0(1-\eta)}$. The second derivative is positive, so it is the minimum. Dividing q^* by $1 - q^* = \frac{w_0(1-\eta)}{w_1\eta + w_0(1-\eta)}$ gives the odds form.

(e) $\frac{\eta}{1-\eta} = \frac{w_0}{w_1} \cdot \frac{q}{1-q} = \frac{1}{19} \times 1.5 = 0.0789$, so $\eta = 0.0789/1.0789 = 0.073$. The weighted model said 0.6; the flow is a 7% case. This is the prior-shift correction: weighting positives by w_1/w_0 is the same as training on a population whose prior odds are multiplied by w_1/w_0 (here $0.05/0.95 \times 19 = 1$, a balanced world), and undoing it rescales the odds by the inverse ratio, exactly $\frac{\pi_d/(1-\pi_d)}{\pi_t/(1-\pi_t)}$ with $\pi_t = 0.5$, $\pi_d = 0.05$.

3. Shrinkage and validation size

(a) $\mathbb{E}\hat{\mu}_c = c\mu$, so bias $= (c-1)\mu$. $\mathrm{Var}(\hat{\mu}_c) = c^2\sigma^2/n$. $\mathrm{MSE}(c) = (1-c)^2\mu^2 + c^2\sigma^2/n$.
(b) $\frac{d}{dc}\mathrm{MSE} = -2(1-c)\mu^2 + 2c\sigma^2/n = 0 \Rightarrow c^* = \frac{\mu^2}{\mu^2+\sigma^2/n}$ (second derivative $2\mu^2 + 2\sigma^2/n > 0$). $c^* < 1$ because the denominator exceeds the numerator when $\sigma^2 > 0$: *the unbiased estimator is never MSE-optimal*; accepting a little bias buys a larger cut in variance. $c^* \to 1$ as $n \to \infty$ because the variance term vanishes: the optimal amount of regularisation shrinks as data grows, the capacity of a model is judged relative to n.
(c) $c = 1$: MSE $= 16/4 = 4$. $c^* = 4/(4+4) = 0.5$: MSE $= 0.25 \times 4 + 0.25 \times 4 = 2$, half. In practice c is a hyperparameter chosen on validation data, as λ is; ridge regression is exactly this: in one dimension $\hat{w}_\lambda = \frac{\sum xy}{\sum x^2 + N\lambda}$ is the least-squares slope multiplied by $\frac{\sum x^2}{\sum x^2 + N\lambda} < 1$.
(d) $2e^{-2n\varepsilon^2} \le \delta \iff n \ge \frac{\ln(2/\delta)}{2\varepsilon^2} = \frac{\ln 40}{2\times 10^{-4}} = 18{,}444.4$, so $n = 18{,}445$.
(e) Let A_j be the event that configuration j is off by more than ε. $\mathbb{P}(\text{some } A_j) \le \sum_j \mathbb{P}(A_j) \le 2m\,e^{-2n\varepsilon^2}$, so all are within ε with probability at least $1 - 2m\,e^{-2n\varepsilon^2}$. Requiring this $\ge 1-\delta$: $n \ge \frac{\ln(2m/\delta)}{2\varepsilon^2} = \frac{\ln 4000}{2\times 10^{-4}} = 41{,}471$. When all are within ε, the selected one is too, so its reported error is honest to $\pm\varepsilon$. This is the winner's curse made quantitative: comparing m candidates costs a factor $\ln m$ in data, and a validation set sized for one model is too small once it is used to choose.
(f) Independence fails: 200 near-copies carry about the information of one example. The effective n is close to the number of episodes, so the true deviation is much larger than the formula with the row count suggests; the guarantee is worse. Hence grouped splits and counting episodes, not rows.

4. Poisson counts and separable data

(a) $\log L(\lambda_c) = \sum_{i:y=c}[k_i \log\lambda_c - \lambda_c - \log k_i!]$. Derivative: $\frac{\sum k_i}{\lambda_c} - N_c = 0 \Rightarrow \hat{\lambda}_c = \frac{1}{N_c}\sum_{i:y=c} k_i$, the class mean (second derivative $-\sum k_i/\lambda^2 < 0$).
(b) $\log \frac{\pi_1\lambda_1^k e^{-\lambda_1}/k!}{\pi_0\lambda_0^k e^{-\lambda_0}/k!} = \log\frac{\pi_1}{\pi_0} + k\log\frac{\lambda_1}{\lambda_0} - (\lambda_1 - \lambda_0)$; the $k!$ cancels, leaving an affine function of k. Zero at $k^* = \frac{\lambda_1 - \lambda_0 + \log(\pi_0/\pi_1)}{\log(\lambda_1/\lambda_0)}$; alert for $k > k^*$ when $\lambda_1 > \lambda_0$.
(c) $k^* = \frac{5.5+\ln 49}{\ln 12} = \frac{5.5+3.892}{2.485} = 3.78$: alert at $k \ge 4$. Log-odds at $k=3$: $-3.892 + 7.455 - 5.5 = -1.937$, posterior 0.126. At $k = 4$: 0.548, posterior 0.634. One extra failed login moves the posterior from 13% to 63%: the likelihood ratio per login is $\lambda_1/\lambda_0 = 12$.
(d) $\mathbb{P}(y=1 \mid k) = \sigma(wk+b)$ with $w = \ln 12 = 2.485$ and $b = \ln(0.02/0.98) - 5.5 = -9.392$. With 30 attack minutes the generative route needs only one number per class (a mean) and estimates it well; logistic regression must find the same two numbers from the few positives by optimisation and is noisier. With bursty attacks the Poisson model is misspecified (variance $\gg$ mean): its likelihood ratio is wrong and its posteriors are badly calibrated, while logistic regression only assumes the log-odds is linear in k, which can still hold. The generative route degrades first.
(e) Write $s_i = w^\top x^{(i)}$. For $y^{(i)} = 1$ the term is $-\log\sigma(as_i)$ with $s_i > 0$: $\sigma(as_i)$ increases strictly to 1 as a grows, so the term decreases strictly to 0. For $y^{(i)} = 0$ the term is $-\log(1-\sigma(as_i)) = -\log\sigma(-as_i)$ with $-s_i > 0$: same conclusion. The average of strictly decreasing terms tending to 0 is strictly decreasing to 0. The loss is positive for every finite θ and its infimum 0 is approached only as $\|\theta\| \to \infty$: no minimiser. Gradient descent keeps decreasing the loss by growing $\|\theta\|$ without bound, along the separating direction.

(f) $J(\theta) = \widehat{R}_{\mathrm{CE}}(\theta) + \lambda\|\theta\|^2 \geq \lambda\|\theta\|^2 \to \infty$ as $\|\theta\| \to \infty$, and J is continuous, so it attains a minimum (strict convexity also makes it unique). Without the penalty the fitted posteriors are pushed to exactly 0 and 1 on the training data and far beyond it: the analyst sees "attack, probability 1.000" for flows that are merely on the right side of the line, and any cost threshold becomes meaningless.

5. Other boosting losses, few good features

(a) $g = -y\,e^{-yF}$, $h = y^2 e^{-yF} = e^{-yF}$ since $y^2 = 1$.
(b) $\mathbb{E}[e^{-YF} \mid x] = \eta e^{-F} + (1-\eta)e^{F}$. Derivative: $-\eta e^{-F} + (1-\eta)e^{F} = 0 \Rightarrow e^{2F} = \frac{\eta}{1-\eta} \Rightarrow F^* = \frac{1}{2}\log\frac{\eta}{1-\eta}$ (second derivative positive). The score estimates *half* the log-odds, so $\hat{\eta} = \sigma(2F) = \frac{1}{1+e^{-2F}}$.
(c) $w = -\frac{\sum g_i}{\sum h_i} = \frac{\sum_i y_i a_i}{\sum_i a_i}$ with $a_i = e^{-y_i F_i}$: a weighted average of the labels, each example weighted by how badly the current ensemble treats it ($a_i > 1$ when $y_i F_i < 0$). This is the reweighting view of AdaBoost. Leaf: $a = (1, 1, e^3) = (1, 1, 20.09)$, $w = \frac{1+1-20.09}{22.09} = -0.82$. The one negative example scored $+3$ outweighs the two positives ten to one and drags the leaf negative. Under the log-loss its gradient is $|p - y| = \sigma(3) = 0.95$, bounded by 1, against weight 20 here and $e^{|F|}$ in general. With 10% wrong labels the mislabelled examples are exactly the confidently "wrong" ones, and the exponential loss lets them dominate; use the log-loss.
(d) $\ell = |y - F|$: $-\partial\ell/\partial F = \operatorname{sign}(y - F)$, the pseudo-residuals are ± 1. $h = 0$ almost everywhere, so $-G/H$ is undefined. Fit the tree to the signs, then choose each leaf value by $\min_w \sum_{i\in q} |r_i - w|$; this is the absolute-loss point prediction of Lecture 2, the median of the residuals in the leaf. For $(-3, -1, 0.5, 10)$: any $w \in [-1, 0.5]$ (convention: -0.25); the mean is 1.625, pulled by the single 10. Robustness: one extreme residual (a mislabelled flow, a sensor glitch) cannot move the fit.
(e) The m_{try} candidates are a uniformly random subset. No informative feature is drawn iff all candidates come from the $d - k$ others: $\binom{d-k}{m}/\binom{d}{m}$. Expanding the binomials, $\frac{(d-k)!\,(d-m)!}{(d-k-m)!\,d!} = \prod_{i=0}^{k-1} \frac{d-m-i}{d-i}$. For $m = 5$: $\frac{25\cdot24\cdot23}{30\cdot29\cdot28} = 0.567$, so 0.433. For $m = 10$: $\frac{20\cdot19\cdot18}{30\cdot29\cdot28} = 0.281$, so 0.719. With $\sqrt{30} \approx 5$, more than half the splits offer only noise features and the tree splits on noise: individual trees become too weak for decorrelation to pay. When informative features are rare, m_{try} must be larger; tune it on OOB error, do not take the default.

6. From one flow to one host

(a) $P = \mathbb{P}(Y = 1 \mid \text{alert}) = \frac{\mathbb{P}(\text{alert}|1)\pi}{\mathbb{P}(\text{alert}|1)\pi + \mathbb{P}(\text{alert}|0)(1-\pi)} = \frac{\pi R}{\pi R + (1-\pi)\mathrm{FPR}}$ (Bayes' rule). $\pi = 0.1$: $0.09/(0.09 + 0.009) = 0.909$. $\pi = 0.001$: $0.0009/(0.0009 + 0.00999) = 0.083$. R and FPR are conditional on the class, so under prior shift (same $p(x \mid y)$) they do not change; precision mixes them with π and collapses with the base rate: eleven of every twelve alerts are false at $\pi = 0.001$, with the same detector.
(b) Under independence given the hypothesis, the likelihood of a sequence with m alerts is $0.3^m 0.7^{n-m}$ under H_1 and $0.02^m 0.98^{n-m}$ under H_0; the posterior odds are prior odds times their ratio; take logs. Numbers: $-6.907 + 2.708m - 0.3365(20 - m) > 0 \iff 3.045m > 13.636 \iff m > 4.48$. So $m = 5$ (log-odds $+1.59$, posterior 0.83); $m = 4$ gives -1.46.
(c) Naive Bayes over flows: the flows play the role of features, assumed conditionally independent given the host's state. With bursts of three alerts per event, one event is counted three times: its log-likelihood ratio 2.708 is added three times instead of once, so five alerts may be fewer than two real events and the posterior is badly overconfident (the duplicated-feature failure). Count events, not alerts, or model the burst.

Tools used: conditional risk, total probability, Bayes' rule, one derivative, Gibbs, a union bound, counting subsets. Nothing else.